Janu Kumar (J.K.) Sah

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Research Interests

Fluid–structure interaction and scientific machine learning, through experiments and computation. A question that runs through my work: once every cost is counted, does a new method actually beat a well-tuned classical one?

- **Scientific machine learning for mechanics:** neural operators (DeepONet, FNO) benchmarked against FEM and reduced-order models; phase drift in neural surrogates of oscillatory flows; spiking networks for low-energy sensing.
- **Experimental fluid–structure interaction and bio-inspired sensing:** seal-whisker and lateral-line-inspired sensor arrays, embedded data acquisition, towing-tank experiments, and learned decoders for wake detection and tracking.
- **Sustainable engineering:** life-cycle assessment of CO_2 mineralization in concrete; building and industrial energy audits.

Education

Ph.D., Mechanical Engineering — Jan 2026 – Present
Rochester Institute of Technology, Kate Gleason College of Engineering — *Rochester, NY*

- **Advisor:** Dr. Qian Xue | Flow Physics and Modeling Lab
- **Dissertation:** *From Nature to Prototype: Development of a Bio-Inspired Seal Whisker Sensor Array*
- **Coursework:** Engineering Analysis, Continuum Mechanics, Translating Discovery into Practice, Doctoral Seminar

B.E., Mechanical Engineering — Nov 2018 – Apr 2023
Pulchowk Campus, Institute of Engineering, Tribhuvan University — *Kathmandu, Nepal*

- Recipient of a merit-based full scholarship for undergraduate study.
- **Thesis:** Study of the energy scenario and improvement opportunities of the ICT building, Pulchowk Campus.

Research Experience

Graduate Research Assistant — Jan 2026 – Present
Flow Physics and Modeling Lab, Rochester Institute of Technology — *Rochester, NY*
NRL-funded project on bio-inspired seal-whisker sensor arrays for autonomous underwater detection and tracking (U.S. Naval Research Laboratory; PIs: Dr. Xudong Zheng and Dr. Qian Xue).

- Built and brought up a **32-channel whisker-array data-acquisition system** (24-bit $\Delta\Sigma$ ADCs over SPI); released it as a **Python package** for synchronized logging and visualization.
- Developed **CNN–LSTM models** mapping multi-channel whisker-deflection signals to hydrodynamic motion state for wake-based object localization and tracking.
- Characterized **whisker–wake interaction** in towing-tank experiments and built a frequency-domain pipeline separating whisker self-vibration from wake-induced forcing.

Research Assistant — Apr 2023 – Mar 2024
Center for Energy Studies, Institute of Engineering, Tribhuvan University — *Lalitpur, Nepal*

- Contributed to 25+ solar-monitoring projects and energy audits of industrial sites (Supervisor: Dr. Shree Raj Shakya).
- Research focus: national energy-intensity assessment, wind-energy potential for water pumping, and industrial sustainability-index calculation.

Professional Experience

Mechanical Design Engineer — Apr 2024 – Nov 2025
Ozaki Design Office — *Niihama City, Japan*

- Performed CAD design, 3D modeling, and engineering simulation; conducted industrial site visits to study and analyze mechanical components.

Intern — Apr – May 2022
Alternative Energy Promotion Center — *Kathmandu, Nepal*

- Gained hands-on experience in energy auditing and management, renewable energy, and energy policy, regulation, and outreach.

Publications

Peer-Reviewed Journal Articles

1. **Sah, J. K.** (2026). When is carbonation-cured concrete actually cleaner? A systematic review, a net-benefit condition framework, and a proposed minimum reporting set for CO_2 mineralization in cement-based products. *Journal of Cleaner Production*, 579, 149605. doi:10.1016/j.jclepro.2026.149605
2. Shakya, S. R., **Sah, J. K.**, Bhujel, P., Aryal, H., Bajracharya, T. R., Bhandari, R., Subedi, S., & Chaulagain, B. (2024). Energy intensity improvement potential of the commercial buildings in developing countries: A case study of Nepal. *Journal of Innovations in Engineering Education*, 7(1), 84–95.

3. Shakya, S. R., Subedi, S., Bajracharya, T. R., Shrestha, S. L., **Sah, J. K.**, & Chaulagain, B. (2025). Sustainability Index for a Manufacturing Industry: A Case Study of Cement Industry in Nepal. *Journal of Science and Engineering*, 12(1), 37–48.
4. Shakya, S. R., Chaulagain, B., Poudel, S., Subedi, T. R., Poudel, P. S., Subedi, S., & **Sah, J. K.** (2025). Computation and Experimental Analysis of Helical Bach-Type Savonius Wind Turbine for Water Pumping Application. *Journal of the Institute of Engineering*, 18(1), 94–101.
5. Shah, M. P., & **Sah, J. K.** (2023). Design Optimization of Fins of a Sounding Rocket for Maximum Lift-to-Drag Ratio and Minimum Radar Cross-Section Area Using ANSYS. *International Journal on Engineering Technology*, 1(1), 53–61.

Manuscripts Under Revision

1. **Sah, J. K.** Do neural operators beat classical solvers for linear elasticity? A strong-baseline, full-cost benchmark. *Computational Mechanics* (Springer), under revision.

Peer-Reviewed Conference Proceedings

1. **Sah, J. K.**, Ghimire, I., Tamang, B., & Adhikari, N. (2025). CFX Analysis to Study the Effects of Blade Exit Angle on Performance of Centrifugal Pumps. In *Vibration Engineering and Technology of Machinery, Vol. I (VETOMAC 2022)*, Lecture Notes in Mechanical Engineering. Springer, Singapore.
2. **Sah, J. K.**, & Paudel, A. (2023). Cost Analysis of Owning Four-Wheel Electric Vehicles in Nepal. *International Conference on Engineering & Technology*, Vol. 5, KEC, 112–116.

Technical Skills

- **Programming & ML:** Python (PyTorch, NumPy, SciPy, scikit-learn), MATLAB; physics-informed neural networks, DeepONet / neural operators, reinforcement learning (SAC).
- **Simulation & Modeling:** Fluid–structure interaction, CFD, immersed-boundary methods; ANSYS (CFX/Fluent).
- **Embedded & Instrumentation:** SPI sensor interfacing, 24-bit $\Delta\Sigma$ ADC bring-up, PCB debugging, oscilloscope / multimeter benchtop testing, DAQ system design.
- **CAD & Engineering Tools:** SOLIDWORKS, AutoCAD, SketchUp; EnergyPlus, LEAP, PVsyst (energy modeling).
- **General:** LaTeX, Git, Linux; experimental design, data analysis, and technical writing.

Selected Projects

- **Whisker-Sensor Autonomous Underwater Vehicle — Continuity of RIT MSD P25937.** Continuing an undergraduate senior-design handoff (32-channel acquisition PCB, whisker array, and shroud prototype); developing the Python logging/visualization package and advancing the remaining sensing-to-autonomous-navigation work toward a functional underwater vehicle.
- **Physics-Informed DeepONet for 2D Linear Elasticity.** Developed an operator-learning surrogate predicting displacement/stress fields for static linear elasticity, packaged as a standalone study.
- **Industrial Energy Audit — Dairy Development Corporation, Kathmandu.** Proposed an integrated recovery system: 138 KVA/month demand reduction via power-factor correction (0.67→0.98) and 15% boiler-fuel savings through insulated cladding (2-month payback).
- **EV Cost-Competitiveness Model (Kathmandu Valley).** MATLAB lifetime-cost model showing 40% lower cost for converted EVs vs. new ICE vehicles at 10,000 km/yr.

Talks and Presentations

- **Oral Presentation** — "From Nature to Prototype: Development of a Bio-Inspired Seal Whisker Sensor Array," 2026 RIT Graduate Showcase. Apr 2026
- **Exhibitor** — "From Nature to Prototype: Seal Whisker Sensors," Imagine RIT. Apr 2026
- **Poster** — "Effects of blade exit angle on the performance of a centrifugal pump," VETOMAC 2022, Tribhuvan University, Nepal. Dec 2022

Honors and Awards

RIT Ph.D. Merit Scholarship / Graduate Assistantship — Jan 2026 – Present

Merit-Based Full Scholarship for Undergraduate Study — Nov 2018 – Apr 2023

Professional Service, Leadership & Outreach

- **Reviewer**, *Journal of Building Engineering* (Elsevier).
- **President**, Saraswati Pooja Committee, NTBNS, IOE Pulchowk Campus (2022–2023).
- **Exhibition Coordinator**, 13th National Mechanical and Aerospace Engineering Exhibition, SOMAES (2022–2023).